

Smart Campus AI System

Minimum Viable Product (MVP) Document

1. Introduction

The Smart Campus AI System is a web-based platform developed to streamline communication between students and campus administration. The system provides instant AI-powered assistance, digital complaint management, role-based administration, and complaint analytics within a single platform.

The Minimum Viable Product (MVP) represents the first functional version of the system that delivers core value to students and administrators while validating the effectiveness of the solution.

2. MVP Objective

The primary objective of the MVP is to create a centralized platform that allows students to receive instant support, submit complaints digitally, track complaint progress, and enable administrative staff to manage and resolve issues efficiently.

The MVP focuses on solving the most critical problems faced by students and administrative departments while keeping implementation simple and scalable.

3. Target Users

Students: Students require a simple and efficient way to:

- Ask campus-related questions
- Submit complaints
- Track complaint resolution status

Administrative Staff: Administrative users require:

- Access to complaint records
- Status management capabilities
- Department-specific complaint visibility

Management: College management requires:

- Complaint analytics
- Resolution monitoring
- Performance tracking

4. MVP Features

Feature 1: AI Chatbot

Description

An intelligent chatbot that provides instant responses to common student queries.

Supported Query Categories

- Fee Information
- Examination Details
- Timetable Information
- Attendance Queries

Expected Outcome

Students receive immediate answers without needing to contact administrative staff.

Feature 2: Complaint Submission System

Description

Students can submit complaints through a digital form.

Complaint Information

- Student Name
- Student Email
- Complaint Category
- Complaint Description
- Submission Date

Expected Outcome

Complaints are recorded digitally and stored securely within the system.

Feature 3: Complaint ID Generation

Description

Upon successful submission, the system automatically generates a unique complaint identifier.

Expected Outcome

Each complaint can be uniquely identified and tracked throughout its lifecycle.

Feature 4: Complaint Tracking

Description

Students can track the progress of their complaints using the generated Complaint ID.

Complaint Statuses

- Pending
- Seen
- Resolved
- Delayed

Expected Outcome

Students remain informed about complaint progress and resolution timelines.

Feature 5: Role-Based Dashboard

Description

Administrative users receive dashboard access according to their assigned roles.

Roles Included

Admin

- Full system access
- View all complaints
- Manage all complaint statuses

Head of Department (HOD)

- Access department-related complaints

Mentor

- Access student welfare complaints

Training and Placement Officer (TPO)

- Access placement-related complaints

Accounts Department

- Access fee-related complaints

Expected Outcome

Each department handles only relevant complaints, improving efficiency and security.

Feature 6: Complaint Status Management

Description: Authorized users can update complaint status through the dashboard.

Available Status Options

- Pending
- Seen
- Resolved

Expected Outcome: Students receive updated complaint information and administrators can manage workflows effectively.

Feature 7: Analytics Dashboard

Description

A dashboard displaying real-time complaint statistics.

Metrics Displayed

- Total Complaints
- Pending Complaints
- Seen Complaints
- Resolved Complaints

Expected Outcome

Management gains visibility into operational performance and issue resolution trends.

Feature 8: Modern User Interface

Description

A responsive and user-friendly interface designed to improve user experience.

UI Features

- Responsive Design
- Dark Mode Support
- Interactive Components
- Smooth Animations

Expected Outcome

Improved usability and accessibility across devices.

5. MVP Success Criteria

The MVP will be considered successful if:

- Students can successfully submit complaints.
- Students can track complaints using Complaint IDs.
- Administrative users can manage complaint statuses.
- The chatbot answers common campus-related queries.
- Role-based access functions correctly.
- Analytics accurately reflect complaint data.

6. Technical Stack

Frontend

- HTML
- CSS
- JavaScript

Backend

- Python Flask

Database

- SQLite / Relational Database

Deployment

- Render

Version Control

- Git
- GitHub

7. Future Enhancements Beyond MVP

- AI-powered complaint categorization
- Email and SMS notifications
- Automated complaint routing
- Voice-enabled chatbot
- Mobile application
- WhatsApp integration
- Generative AI support
- Predictive analytics
- Sentiment analysis

Conclusion

The Smart Campus AI System MVP delivers the essential functionality required to digitize student support services. It establishes a foundation for future AI-driven campus automation while providing immediate value through complaint management, student assistance, and administrative oversight.